

Energy-system interactions reshape offshore hydrogen trade and carbon mitigation in Europe

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Abstract

Offshore wind-rich island systems are often assessed through production costs, but their hydrogen export value also depends on domestic absorption, shipping, demand geography and competing imports. Here we present the first integrated European-scale modelling framework linking offshore hydrogen cost-supply curves, hourly power-gas operation, cross-border hydrogen/ammonia shipping and European demand from 2030 to 2050. A lowest-LCOH-only allocation identifies Denmark or the Netherlands as leading exporters, whereas the integrated model identifies the UK as the leading net exporter in all three model years, with Ireland nearly matching it by 2050. In the integrated 2050 case, the UK and Ireland each reach approximately 162 TWh yr⁻¹ of net exports, and the full supply mix mitigates about 175.5 Mt CO₂ yr⁻¹ in the modelled displacement accounting, of which 158.3 Mt CO₂ yr⁻¹ is attributed to European offshore suppliers. The result is conditional on the external import option: delivered outside prices near 2–3 EUR kg⁻¹ H₂-equivalent displace most European offshore supply, whereas prices around and above 4 EUR kg⁻¹ restore the UK/Ireland export pattern. These findings show that strategic offshore hydrogen exporters cannot be inferred from LCOH alone.

Keywords: hydrogen, offshore wind, carbon mitigation, international trade

1 Introduction

Europe’s offshore wind expansion creates a spatial allocation problem, not only a generation-cost problem. Ireland and the UK have announced offshore wind development plans reaching 37 and 125 GW of installed capacity, respectively, by 2050 [1, 2]. These targets are more than twice the projected values of their total power demands in 2050 [3, 4]. Connecting and using such large offshore generation volumes through domestic power systems alone is challenging because domestic demand is limited and real-time balancing requirements increase with uncertain and intermittent offshore wind.

Green hydrogen offers a route for moving offshore renewable energy across sectors and borders. Renewable electrolysis avoids the direct operational emissions associated with grey hydrogen and can reduce lifecycle emissions relative to fossil-based production routes [5]. Hydrogen can also be stored and transported at larger scales than electricity, allowing offshore wind output to serve domestic industry, power and heat demands or distant demand centres through pipelines and shipping. This matters because prospective exporters such as Ireland and the UK sit near large future import needs: Germany expects annual imported hydrogen needs of 45–77 TWh by 2030 [6, 7], and Belgium has identified 20 TWh of imported hydrogen demand by 2030 [8].

Production cost remains central to offshore hydrogen deployment, but it is not sufficient for identifying strategic exporters. The levelised cost of hydrogen (LCOH) in 2030 has been projected at around 54–184 €/MWh, overlapping with blue hydrogen with carbon capture and storage and green hydrogen from onshore wind [9]. Country- and region-specific LCOH studies have assessed North Sea countries [10], the UK [11], Poland [12] and the Iberian Peninsula [13, 14], often with detailed cost breakdowns, production routes and uncertainty analysis. However, separate national studies cannot show how production costs interact with domestic absorption, shipping costs, European demand geography and competing imports.

This gap is particularly important for Ireland and the UK. European hydrogen strategies often emphasise imports from North Sea countries, North Africa, the Middle East and the Americas, while the export potential of these Atlantic-facing island systems is less quantified. Policy and consulting reports have identified opportunities for Irish and UK hydrogen exports [1, 15], but a pipeline-only or LCOH-only view can miss how their offshore resource value changes once domestic energy-system constraints and European trade balances are considered together. The missing element is not another standalone production-cost estimate; it is a coupled assessment of where offshore hydrogen can be produced, how much can be absorbed domestically, how surplus can move through hydrogen and ammonia trade, and how competing imports affect the resulting carbon-mitigation value (Supplementary Table 1).

This work provides the first integrated European-scale assessment of offshore hydrogen trade that links geospatial production costs, hourly domestic power-gas absorption, shipping-based hydrogen and ammonia exchange, competing external imports, an LCOH-only counterfactual and supplier-specific carbon-mitigation attribution in one optimisation framework. We first construct geospatial cost-supply curves for offshore hydrogen in 11 coastal countries using meteorological, bathymetric, infrastructure and exclusion-area data. We then use an hourly coordinated

power-gas optimisation model for Ireland to estimate how domestic energy systems absorb offshore wind and hydrogen under evolving operating constraints, and extend the resulting export-potential logic to the wider country set. Finally, we combine cost-supply curves, domestic export limits, shipping costs, national hydrogen and ammonia demand, and an external import option in an international trade optimisation model. This structure lets us test whether strategic exporters can be inferred from LCOH alone and how robust the resulting trade geography is to competing imported hydrogen.

2 Results

2.1 Offshore hydrogen costs do not determine export value alone

We assess Belgium, Denmark, France, Germany, Ireland, the Netherlands, Norway, Portugal, Spain, Sweden and the United Kingdom (UK), which combine large exclusive economic zones (EEZs), offshore infrastructure potential and navigable links to European demand centres. Wind velocity, air pressure and temperature data are obtained from ERA5 at 1 hour and $0.5^\circ \times 0.5^\circ$ temporal and spatial resolutions [16]. Hourly capacity factors and wake-effect losses are simulated using a 15 MW reference wind turbine [17]. The resulting wind-speed distributions show that Ireland has the highest modal wind speed among the studied EEZs, followed by the UK, mainly because of their exposure to Atlantic wind resources (Fig. 1b).

We then combine these capacity factors with development, capital, operating and decommissioning costs for offshore hydrogen production platforms, including turbines, foundations, electrolysers and auxiliary systems (Methods). Bathymetry, port locations and power/gas connection points are included so that water depth, installation distance and maintenance distance affect site-level costs [18, 19]. The resulting LCOH ranges from 84.93 to more than 200 €/MWh across the studied EEZs (Fig. 1a). Denmark has the lowest-cost locations because it combines high capacity factors with a shallow continental shelf, although the lower-bound offshore hydrogen cost remains 33.94% higher than an onshore blue-hydrogen benchmark of 68.92 €/MWh [20]. The UK’s LCOH ranges from 85.97 to 307.26 €/MWh, while Ireland’s ranges from 90.42 to 201.8 €/MWh. These results show that Atlantic wind-resource quality does not automatically translate into the lowest production cost once foundation depth and infrastructure distances are included.

We further account for wake losses, vessel density, excluded areas such as military and aquaculture zones, and future cost reductions [21] to construct country-level cost-supply curves for 2030, 2040 and 2050 (Fig. 2). In 2030, marginal LCOH is 76.10 €/MWh in Ireland and 79.02 €/MWh in the UK. Ireland is slightly cheaper at the policy-constrained marginal capacity, but the UK retains more low-cost near-shore supply. Denmark remains the leading cost benchmark, while France and the Netherlands retain competitive supply segments even when their full offshore capacity is not the lowest-cost option.

Technology and supply-chain learning reduce LCOH in later scenarios, but they do not make the Atlantic island systems simple lowest-cost winners. By 2050, marginal

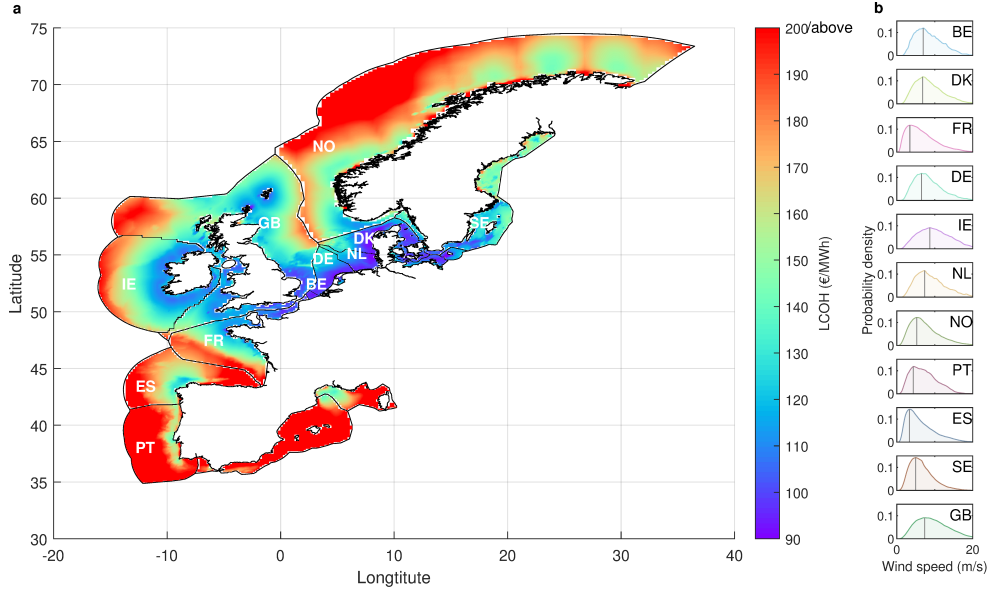


Fig. 1 LCOH map and wind speed in coastal European countries. **a** LCOH map in EEZs of coastal European countries. The black line outlines the EEZs. **b** Probability density of wind speed for each country. The vertical grey line in each figure shows the wind speed most likely to occur. BE, DK, FR, DE, IE, NL, NO, PT, ES, SE, and GB are the abbreviations for Belgium, Denmark, France, Germany, Ireland, The Netherlands, Norway, Portugal, Spain, Sweden, and the United Kingdom, respectively.

LCOH falls to 54.42 €/MWh for Ireland and 52.51 €/MWh for the UK, and all studied countries have offshore hydrogen segments that can compete with the assumed blue-hydrogen cost range (Fig. 3). The key implication is that Ireland and the UK have sufficient cost competitiveness to participate in offshore hydrogen trade, but their strategic value must be tested against domestic absorption, shipping and demand.

2.2 Domestic power-gas constraints create exportable offshore surplus

Because the offshore wind ambitions of Ireland and the UK exceed their projected domestic power demands, we next estimate how much offshore production can be absorbed domestically before it becomes available for export. We develop a whole-energy-system optimisation model that simulates hourly power and gas operation over a full year under uncertain offshore wind generation. Power and natural gas systems are coupled through gas-fired power plants and onshore/offshore electrolyzers, allowing bidirectional energy conversion to expand system flexibility. Hydrogen blending and pure-hydrogen network cases are represented through different maximum hydrogen fractions in the gas network. Hourly unit commitment includes the All-Ireland generator fleet, interconnectors, reserve, inertia, must-run and system non-synchronous penetration constraints, and observed hourly demand profiles (Methods; Supplementary Information) [22].

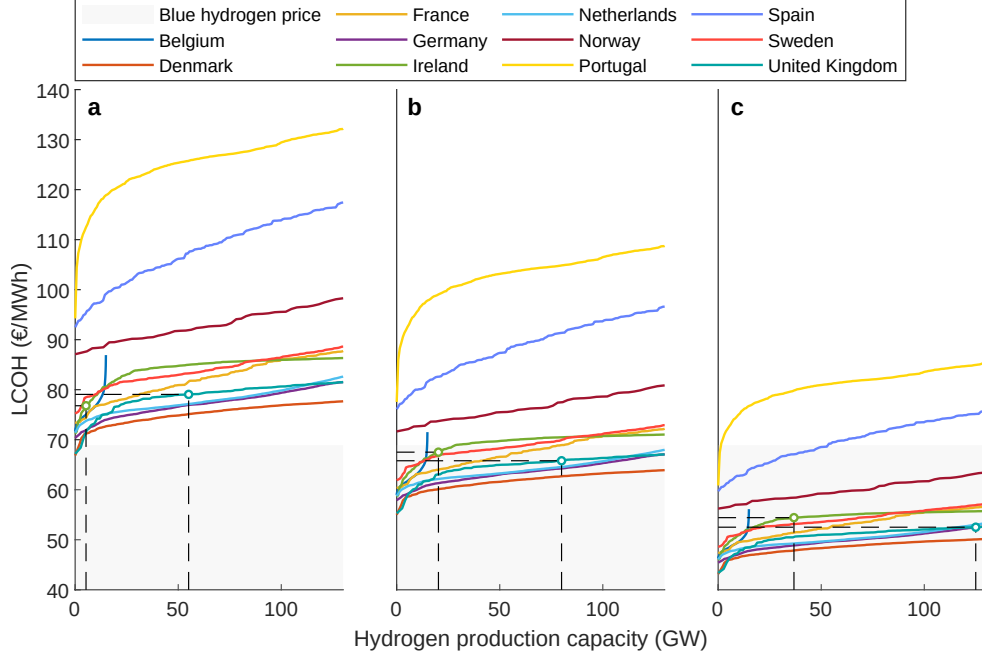


Fig. 2 Hydrogen cost-supply curves for different countries at different years. The cost-supply curves of green hydrogen produced by countries in 2030, 2040, and 2050 are shown in **a**, **b**, and **c**, respectively. Different colours of solid lines represent different countries. Green and blue dots on Ireland's and the UK's curves mark their goals for offshore wind capacities. Grey areas represent the possible range of blue hydrogen production prices in the future.

The domestic absorption summary compares seasons, model years and export/no-export cases (Fig. 4). We assume a maximum 20% hydrogen blending ratio in 2040, reflecting near-term blending demonstrations and policy movement, and a pure-hydrogen network in 2050 in line with the UK's hydrogen network roadmap [23]. Without export, the average summer wind-curtailment rate increases from 62.36% in 2030 to 69.80% in 2050 because high offshore generation coincides with low seasonal demand. In winter, the curtailment rate falls from 32.05% in 2030 to 13.47% in 2050 because higher hydrogen blending allows the gas system to absorb more offshore energy.

Even after extensive hydrogen use in the gas system, domestic demand cannot absorb the full offshore wind output. Export access substantially increases utilisation across all model years and seasons by creating an outlet for surplus hydrogen. In the 2050 summer case, for example, the average wind-curtailment rate falls to 10.46% with export access, and the full-year decomposition identifies approximately 170 TWh yr^{-1} of Irish export potential. Curtailment does not fall to zero because operating reserves, inertia requirements and must-run constraints still limit how much uncertain offshore output can be used at the margin.

We then use the full-year Ireland results to estimate offshore wind utilisation for the wider country set, assuming that domestic offshore absorption scales with

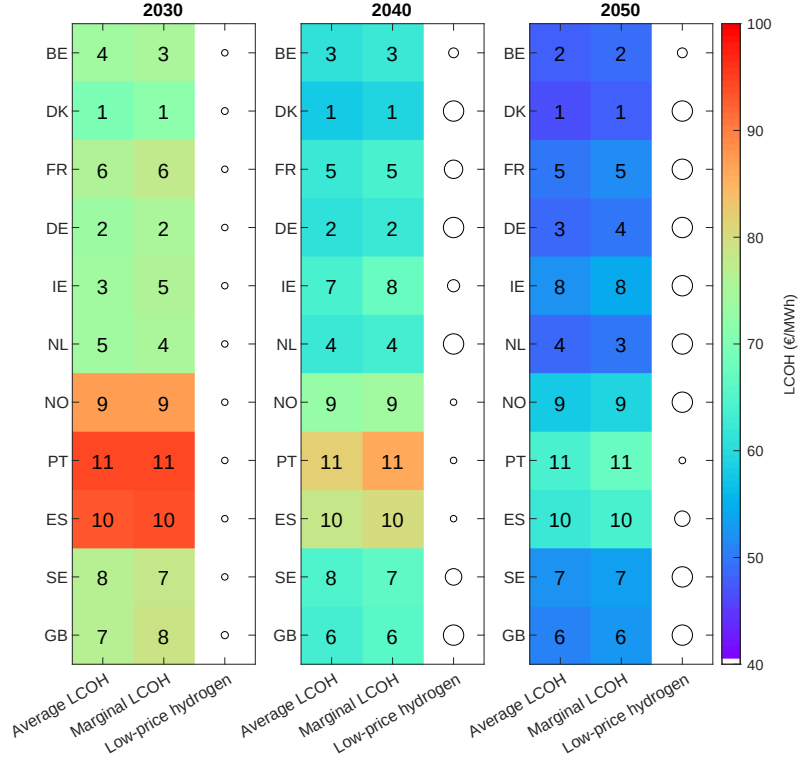


Fig. 3 Cost-competitiveness of offshore green hydrogen among major countries and with blue hydrogen. Rank indicates the average or marginal LCOH rank among the 11 studied countries. Marginal means the LCOH at the policy-constrained offshore capacity of that country in the corresponding year. Average means the capacity-weighted average LCOH across the offshore capacity in the corresponding year. Low-price hydrogen means the hydrogen production capacity with production cost below the upper bound of blue hydrogen. The size of the circle indicates the capacity of low-priced hydrogen. The smallest and largest circle sizes indicate 0 and >100 GW, and intermediate sizes indicate capacities between these values.

electricity and gas demand. This is a screening-level extrapolation, not a substitute for country-specific unit-commitment studies; we therefore test its effect through a domestic-absorption sensitivity in the Supplementary Information. The annual decomposition identifies available export potential rather than optimised net trade (Fig. 5). In 2030, the UK has the largest available export potential at 176 TWh yr^{-1} because its offshore capacity is large relative to domestic load. The Netherlands, Denmark and Ireland follow with 64, 30 and 16 TWh yr^{-1} , respectively. In 2040, the UK and the Netherlands remain the two largest potential exporters, with 195 and 155 TWh yr^{-1} , while Ireland reaches 87 TWh yr^{-1} and becomes the third-largest potential exporter. By 2050, the UK and Ireland reach 263 and 170 TWh yr^{-1} , respectively, while the Netherlands' export potential falls because higher domestic hydrogen use absorbs more offshore output. Ireland therefore becomes the second-largest source of available export potential, despite not being a lowest-LCOH country.

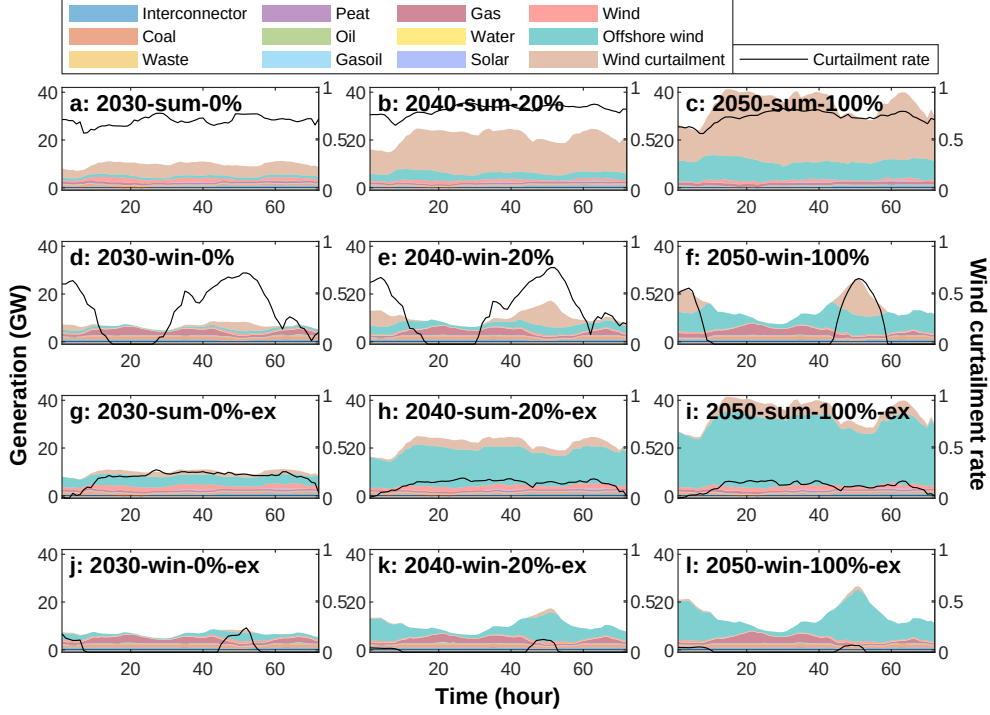


Fig. 4 Export access changes Ireland's hourly power-gas operation by reducing offshore wind curtailment. Panels compare representative 72-hour summer (sum) and winter (win) periods in 2030, 2040 and 2050. The first two rows show cases without export access, and the last two rows show otherwise matched cases with export access (ex). Coloured areas show interconnector power and generation by technology, with offshore wind and curtailed offshore wind separated from other generation. Black lines show the offshore wind curtailment rate on the right axis. The 0%, 20% and 100% labels denote the allowed hydrogen share in the gas network, with 100% representing pure-hydrogen network operation.

2.3 Integrated trade reallocates carbon-mitigation value across Europe

We next combine the cost-supply curves and export-potential estimates in an international green hydrogen trading model for 2030, 2040 and 2050. The cost-supply curves are interpolated as piecewise linear functions and combined with shipping costs, national hydrogen and ammonia demand, domestic export limits and external imports in a cost-minimising optimisation. The resulting flows and supplier contributions to carbon mitigation are shown in Fig. 6.

In 2030, the UK is the major hydrogen exporter, followed by the Netherlands, Denmark, and Ireland, with net exports of approximately 78, 39, 27 and 14 TWh yr⁻¹, respectively. Germany remains the major hydrogen receiver due to its high demand, so the lowest-cost and geographically close suppliers do not determine the whole trade pattern by themselves. By 2040, external hydrogen imports become necessary because offshore capacity growth is slower than demand growth. The integrated solution identifies approximately 116 TWh yr⁻¹ of external imports, while the UK, the Netherlands

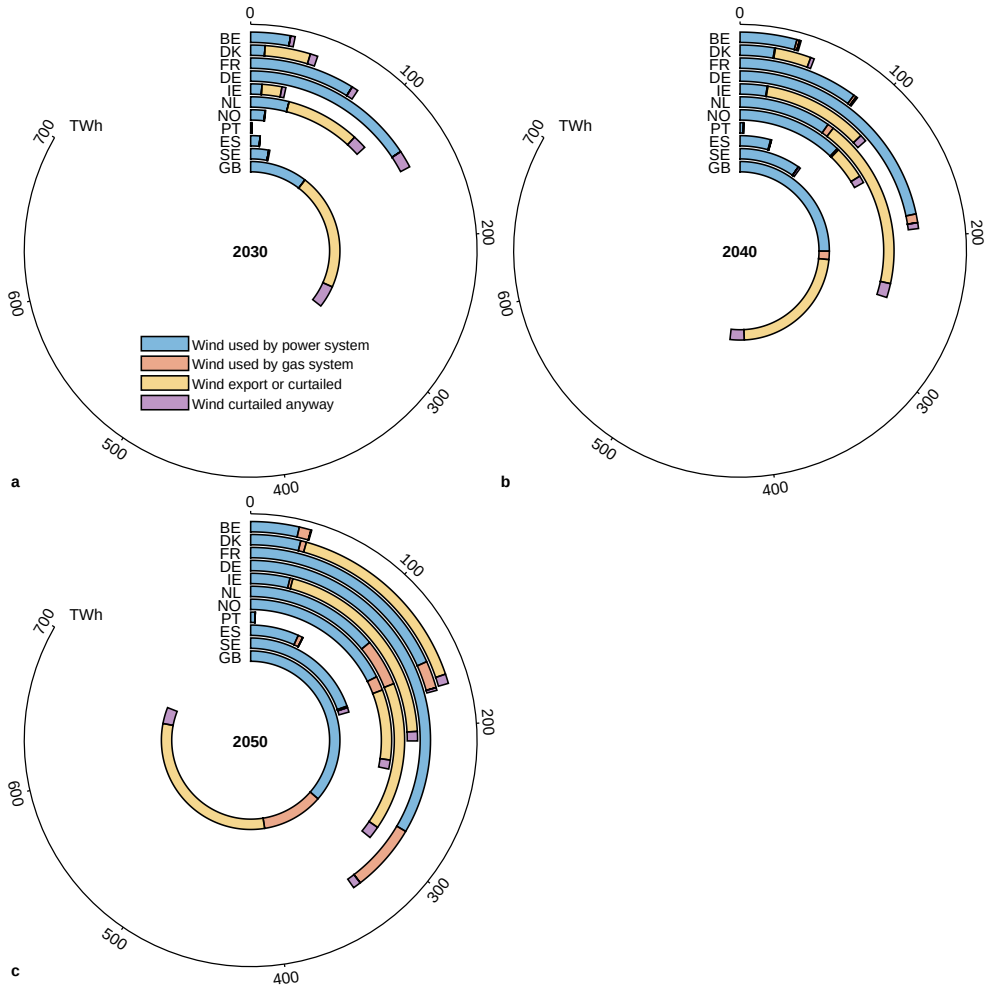


Fig. 5 Europe-wide offshore wind utilisation identifies exportable surplus after domestic power-gas absorption. a–c, Decomposition of annual offshore wind use across the 11 studied countries in 2030, 2040 and 2050. Each circular stack separates wind used by the power system, wind used by the gas system, wind available for export or otherwise curtailed, and residual wind curtailed anyway. Radial values are in TWh yr⁻¹. Country abbreviations follow Fig. 1. The panel highlights that the UK remains the largest available-surplus system, while Ireland becomes the second-largest by 2050 despite not being a lowest-LCOH country.

and Ireland remain large European net exporters with approximately 122, 85 and 83 TWh yr⁻¹, respectively. In 2050, external imports fall to approximately 62 TWh yr⁻¹ but remain part of the overall supply structure. The UK and Ireland become the two largest European net exporters at approximately 162 TWh yr⁻¹ each, followed by Denmark at approximately 118 TWh yr⁻¹.

Hydrogen trade also reallocates carbon-mitigation value across Europe (Fig. 6b,d,f). In the modelled displacement accounting, the integrated supply mix mitigates approximately 43.5, 129.8 and 175.5 Mt CO₂ yr⁻¹ in 2030, 2040 and 2050, respectively,

equivalent to 0.74, 2.20 and 2.98 times Ireland's current annual carbon emissions [24]. Of these totals, European offshore suppliers account for approximately 43.5, 103.2 and 158.3 Mt CO₂ yr⁻¹, with the remaining contributions in 2040 and 2050 associated with external imports. In 2030, the UK and the Netherlands are the main European offshore contributors to carbon mitigation. From 2040 to 2050, Ireland's importance increases with its export position. Ireland and the UK therefore mitigate emissions not only through domestic supply, but also through hydrogen delivered to demand centres elsewhere in Europe.

2.4 Production cost alone misidentifies strategic hydrogen exporters

The cost, domestic-absorption and trade results above show that low LCOH is only one part of the offshore hydrogen trade problem. To isolate what is missed by a production-cost view, we constructed a lowest-LCOH-only counterfactual that allocates hydrogen demand to the lowest-cost offshore supply first, and compared it with the integrated model that includes domestic absorption limits, shipping costs, national demand and residual external imports. This counterfactual is not intended as a market-design proposal; it is a diagnostic baseline for testing whether the strategic-exporter ranking can be inferred from LCOH alone.

The two approaches give different exporter geographies (Fig. 7). In the cost-only baseline, the leading net exporters are Denmark in 2030, the Netherlands in 2040 and Denmark again in 2050, with net exports of 44, 147 and 171 TWh yr⁻¹, respectively. In the integrated model, the leading net exporter is the UK in all three model years, with 78, 122 and 162 TWh yr⁻¹, while Ireland nearly matches the UK in 2050 at 162 TWh yr⁻¹. The difference is particularly large in 2050: the cost-only baseline assigns 41 TWh yr⁻¹ of net exports to Ireland and a 44 TWh yr⁻¹ net import position to the UK, whereas the integrated model increases both countries to approximately 162 TWh yr⁻¹ of net exports. The integrated model also reveals residual external import requirements of approximately 116 and 62 TWh yr⁻¹ in 2040 and 2050, while the cost-only baseline hides this shortage by construction. These results show that the policy-relevant exporter ranking is determined by the system balance between cost, domestic absorption, shipping and demand, rather than by LCOH ranking alone.

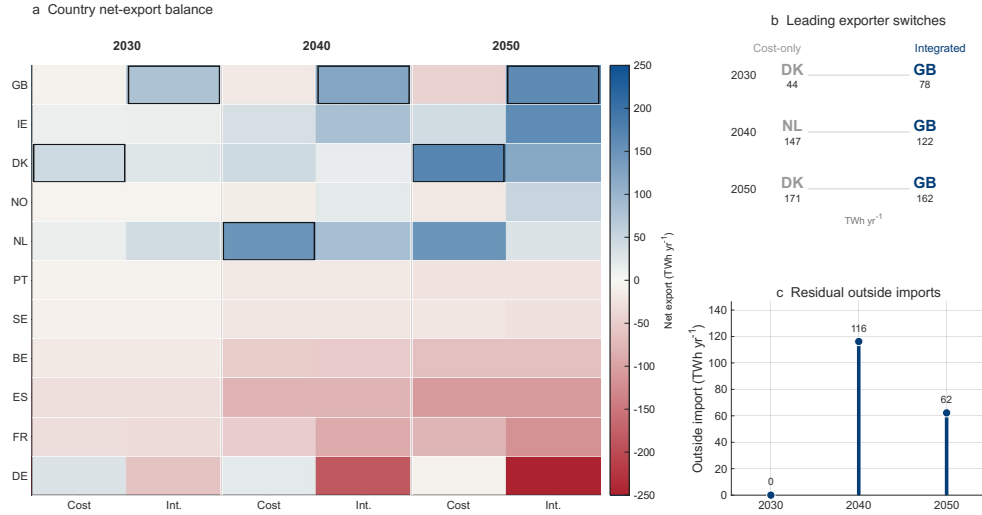


Fig. 7 Lowest-LCOH allocation misidentifies strategic offshore hydrogen exporters. **a** Country net-export balance under a lowest-LCOH-only baseline and the integrated model in 2030, 2040 and 2050. Blue indicates net export and red indicates net import. Black boxes mark the leading net exporter in each model-year combination. **b** Leading net exporter and corresponding net export under the two model structures. **c** Residual outside imports identified by the integrated model. The cost-only baseline allocates demand within the modelled European offshore supply by construction and therefore does not expose the residual external import requirement.

2.5 External import prices define a threshold for European offshore supply

The integrated trade result depends on how attractive non-European low-carbon hydrogen or ammonia imports become relative to European offshore supply. This outside option is strategically important because European policy already anticipates substantial renewable hydrogen imports, while the future scale, timing and cost of global low-emission hydrogen trade remain uncertain [25–27]. We therefore replaced the previous high import penalty with a delivered hydrogen-equivalent outside-option price and solved the trade model over 2–6 EUR kg⁻¹ H₂-equivalent. This range should be read as a threshold scan rather than a price forecast: its low end is informed by optimistic global trade and cost estimates, while the upper range represents less favourable delivery, infrastructure, finance or policy conditions [28, 29].

The outside-option scan shows a threshold rather than unconditional robustness (Fig. 8). In the 2050 case, an external option near 2 EUR kg⁻¹ H₂-equivalent displaces European offshore supply almost entirely, with 818 TWWh yr⁻¹ supplied from outside the modelled European offshore system. At 3 EUR kg⁻¹, European offshore supply remains limited to 98 TWWh yr⁻¹, outside imports remain at approximately 721 TWWh yr⁻¹ and the UK remains a net importer. Around 4 EUR kg⁻¹, the solution changes regime: European offshore supply rises to about 767 TWWh yr⁻¹, outside imports fall to about 63 TWWh yr⁻¹, and the UK, Ireland and Denmark re-emerge as large net exporters, at approximately 162, 161 and 118 TWWh yr⁻¹, respectively. The UK/Ireland

export conclusion is therefore conditional on external imports not being available at ultra-low delivered prices, but it is restored once the outside option enters a more constrained delivered-cost range.

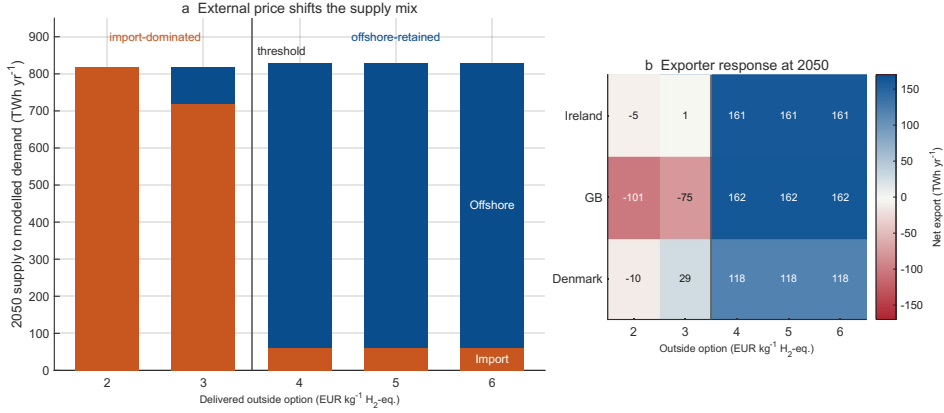


Fig. 8 External import prices define a threshold for European offshore hydrogen supply. **a** 2050 supply mix between external low-carbon imports and European offshore hydrogen under delivered outside-option prices from 2 to 6 EUR kg⁻¹ H₂-equivalent. **b** Net-export response of Ireland, the United Kingdom and Denmark under the same price scan. Blue indicates net export and red indicates net import. The scan is designed to identify a price threshold, not to assign probabilities to future import prices.

3 Discussion

The overall analysis shows that offshore green hydrogen can contribute materially to Europe’s net-zero transition, but its strategic geography is not determined by LCOH alone. Denmark remains the strongest cost-competitiveness benchmark across the cost-supply curves, whereas Ireland and the UK become strategically important only after domestic absorption, national demand, shipping and residual external imports are considered together. In the integrated 2050 case, the UK and Ireland become the two largest net exporters with approximately 162 TWh yr⁻¹ each, followed by Denmark with approximately 118 TWh yr⁻¹.

This distinction matters for policy appraisal. A lowest-cost-only allocation would identify Denmark or the Netherlands as the leading exporters and would hide residual import dependence, whereas the integrated model identifies a west-to-east trade structure with persistent external import needs in 2040 and 2050. The result is therefore not that high-resource island systems automatically dominate offshore hydrogen trade. Rather, their strategic role emerges when large offshore build-out coincides with limited domestic absorption, nearby continental demand and a delivered external import option that is not ultra-low cost.

These findings also clarify how Ireland should be interpreted in future hydrogen planning. Ireland has no decisive LCOH advantage and does not have the largest offshore hydrogen production capacity. Its strategic value arises from the combination of

sufficient cost competitiveness, large exportable surplus and European demand centres that remain undersupplied by their domestic offshore resources. This conclusion supports considering multiple export corridors and carrier options rather than evaluating Irish exports only through a single pipeline route. It also implies that national offshore hydrogen strategies should be assessed within a regional trade and demand balance, not only through domestic cost-supply estimates.

The central innovation of this study is therefore methodological as well as quantitative. By linking geospatial offshore hydrogen cost-supply curves, hourly domestic power-gas absorption, hydrogen/ammonia shipping and an explicit external import option, the model exposes a ranking reversal that single-module assessments cannot capture. The lowest-LCOH baseline points to Denmark or the Netherlands as leading exporters, but the integrated model identifies the UK as the leading exporter in all three model years and Ireland as an almost equal exporter by 2050. This first integrated European-scale treatment is what makes the UK/Ireland result visible.

Several assumptions define the scope of these conclusions. The analysis uses currently published offshore wind goals and therefore assumes that planned infrastructure is delivered on schedule. Domestic absorption outside Ireland is estimated by scaling the Ireland hourly whole-system result with national electricity and gas demand, so the export-potential estimates should be read as a European screening result rather than a replacement for country-specific unit-commitment studies. We stress-test this bridge in the Supplementary Information by resolving the integrated trade model after multiplying domestic absorption by 0.75 and 1.25. This changes European offshore supply, residual imports and the exact UK/Ireland ordering, but it does not restore the LCOH-only exporter geography: Ireland remains a leading net exporter and the ranking reversal relative to the cost-only baseline is preserved. The outside-option analysis is also a threshold scan, not a forecast of future global hydrogen prices. These assumptions do not weaken the integrated trade result, but they identify where country-level planning and future market evidence would most improve the analysis.

The results indicate that residual external imports remain necessary in the integrated 2040 and 2050 cases, at approximately 116 and 62 TWh yr⁻¹, respectively. More ambitious offshore hydrogen production goals could reduce this dependence, especially in countries with lower marginal costs and remaining offshore capacity potential, such as Denmark and the Netherlands. Infrastructure policy also matters. Offshore wind turbines, electrolyzers, subsea cables, ports and hydrogen shipping vessels must be coordinated with stable regulation, faster consenting and dedicated export corridors. Otherwise, European hydrogen demand can grow faster than domestic offshore supply.

These findings give a practical test for future hydrogen strategies. Countries should not be ranked only by LCOH or offshore resource quality. They should be evaluated by their position in the wider system: domestic absorption, exportable surplus, carrier choice, shipping access, demand geography and competing external imports. In that system view, Ireland and the UK become strategic exporters because they are system-positioned, not simply cost-ranked.

4 Methods

4.1 Model workflow and scope

The analysis links four modelling layers. First, geospatial offshore wind, bathymetry, port-distance, grid-connection and exclusion-area data are used to construct country-level offshore hydrogen and ammonia cost-supply curves for 2030, 2040 and 2050. Second, an hourly coordinated power-gas operation model is used to estimate how much offshore wind and hydrogen can be absorbed domestically before an export outlet is required. Third, the resulting export-potential estimates are combined with hydrogen and ammonia demand, shipping costs and an external import node in a European trade optimisation model. Fourth, the optimised supply and delivery matrices are converted into supplier-attributed carbon-mitigation values. The main text reports the model structure and assumptions that determine the results; the mathematical formulations and parameter tables are provided in the Supplementary Information.

The country set comprises Belgium, Denmark, France, Germany, Ireland, the Netherlands, Norway, Portugal, Spain, Sweden and the UK. The trade optimisation is solved separately for 2030, 2040 and 2050. Energy-flow decision variables are solved as average power quantities and converted to TWh yr⁻¹ for reporting. Carbon-mitigation outputs are reported in Mt CO₂ yr⁻¹.

4.2 Techno-economic model of offshore hydrogen production

Offshore wind generation and production cost are assessed within the EEZs of the selected coastal European countries [30]. Aquaculture sites, Special Areas of Conservation and military areas are excluded from the developable offshore hydrogen production area using boundaries from EMODnet [19]. Within the remaining offshore grid cells, hourly wind velocity, temperature and air-pressure data are obtained from the ERA5 reanalysis. ERA5 data are used at 1 h temporal resolution and 0.5° × 0.5° spatial resolution, equivalent to approximately 45 km by 56 km in southern Europe [16].

Wind power generation and capacity factors are calculated hourly using the power curve of an NREL reference 15 MW wind turbine [17]. The turbine power curve is defined at 1 m s⁻¹ wind-speed increments, and linear interpolation is used for intermediate wind speeds. Turbine layouts are optimised by balancing power density against wake-effect losses. The wake effect of a single turbine is simulated using the Jensen-Gaussian model [31]. When multiple upstream turbines affect the same downstream turbine, wake effects can overlap. Enumeration-based approaches become computationally expensive when many turbines and long historical time series are considered [32, 33]. We therefore use a recursive-tree-based method that reduces simulation time by 79% with less than 1% accuracy loss in benchmark tests [34]. The specific methodologies and parameter values are provided in the Supplementary Information.

LCOH is calculated using a bottom-up cost model with development expenditure (DEVEX), capital expenditure (CAPEX), operational expenditure (OPEX) and decommissioning expenditure (DECEX). These cost modules are broken down in the

Supplementary Information. European techno-economic parameters, including environmental survey, wind-turbine and cable costs, are taken from Offshore Renewable Energy Catapult [35].

The geolocation-based LCOH calculation uses five site-specific inputs: water depth, capacity factor, distance to the nearest port, distance to the nearest connection point and turbine spacing. Bathymetry data are obtained from EMODnet and converted to $0.001^\circ \times 0.001^\circ$ grid cells, equivalent to approximately 0.11 km by 0.06 km near the UK [18]. Water depth determines whether monopile, jacket or floating turbine structures are technically applicable and affects foundation material requirements. Depths below 30 m are treated as suitable for monopiles, depths of 30–60 m for jacket foundations, and depths above 60 m for floating foundations. Port distance affects installation and maintenance costs, while distance to connection points affects export-cable CAPEX. Turbine spacing is selected so that wake interference from adjacent turbines remains below 5%, which also determines the array-cable length used in the cost calculation.

The offshore electrolysis platform includes water pumping and desalination, electrolyzers, hydrogen compressors, standby batteries and auxiliary offshore infrastructure. The techno-economic characteristics of alkaline electrolyzers are summarised in Supplementary Table 4. Alkaline electrolysis is used because it is currently a mature and durable electrolyser option for large-scale deployment. Capital-cost and electrical-efficiency projections are taken from Offshore Renewable Energy Catapult and the UK’s National Hydrogen Strategy; alkaline-electrolyser efficiency is assumed to increase from 0.79 in 2030 to 0.82 in 2050 [36]. Annual OPEX includes maintenance, stack replacement and labour costs as a percentage of initial capital investment.

Because offshore wind output is variable and can differ from the electrolyser operating point, electrical and hydrogen storage are included. Water pumping, desalination and compression loads are sized with hydrogen production, so their power consumption scales with hydrogen production capacity. Algebraic sizing equations are used to maximise utilisation and determine the electrolyser and accessory capacities (Supplementary Information, Section 3.3). Future operating parameters and cost projections are taken from Offshore Renewable Energy Catapult [35]. LCOH is then calculated from the DEVEX, CAPEX, OPEX and DECEX of the wind farm, electrolysis system and offshore auxiliaries.

Cost-supply curves are generated by ranking grid-cell production segments by LCOH and accumulating the available hydrogen quantity. The potential installed offshore wind capacity for each grid cell is calculated from turbine rated power and the maximum number of turbines that can be placed in the cell under the wake-interference constraint. Vessel density, obtained from EMODnet at 1 km resolution, reduces available power density linearly up to a maximum 50% reduction [37, 38]. Future LCOH reductions are represented through high, median and low technology-learning scenarios based on expert elicitation [21]. Under the median scenario, relative to 2020, fixed-bottom offshore wind costs are reduced by 21%, 35% and 49% in 2030, 2040 and 2050, respectively; floating offshore wind costs are reduced by 2%, 17% and 40%.

4.3 Domestic offshore wind and hydrogen integration model

Domestic absorption is evaluated with a coordinated operation model for the All-Island power and gas systems. Offshore electricity is connected to the power system by marine cables, while offshore hydrogen can enter the gas system through blending or, in the 2050 case, a pure-hydrogen network assumption. The scenario hydrogen fraction in the gas network is set to 0% in 2030, 20% in 2040 and 100% in 2050. These cases represent a transition from no material blending, through an intermediate blending limit, to a dedicated hydrogen-network case. The 20% case reflects current blending demonstrations and policy discussion [39–42], while the 2050 case is aligned with hydrogen-network planning [43].

The coordinated model determines hourly operation over a representative year while minimising operating cost and maximising offshore-resource integration subject to power-system and gas-system constraints. It is used to estimate domestic offshore wind use, hydrogen production, curtailment and the remaining exportable surplus. Export and no-export cases are solved to distinguish offshore wind that can be absorbed domestically from offshore wind that requires an external hydrogen or ammonia outlet.

The power-system module is a unit-commitment model that schedules resources at nodal resolution to balance electricity demand and uncertain renewable generation. The objective includes thermal-unit fuel costs and start-up costs. The model includes generator ramping limits, minimum online and offline times, minimum output levels, must-run constraints, reserve requirements and the operational policy roadmap for the All-Island system, including inertia, minimum numbers of conventional units and System Non-Synchronous Penetration limits [44]. Network congestion is represented using a DC power-flow model. Seasonal dynamic line ratings are included for summer and winter cases, and planned interconnection expansion, including the 700 MW Celtic Interconnector with France, is represented [45].

The gas-system module minimises gas-system operating cost while accommodating hydrogen injection from offshore electrolysis. Hydrogen blending changes gas composition and therefore changes the heat value, specific gravity and safety characteristics of the gas mixture. We define an admissible gas-quality region using the Gas Safety (Management) (Amendment) Regulations 2023, including relative density, Wobbe index and Weaver flame-speed factor limits [46]. Gas flow and pressure are represented with the Weymouth equation. Because variable gas composition introduces additional non-convexity, the gas-flow problem is solved using the previously developed reformulation framework based on McCormick envelopes, second-order cone reformulation and sequential convex programming [47]. Gas streams with different compositions are mixed at pipeline junctions; the resulting bilinear terms are handled with the same solution framework. Mathematical details are provided in Supplementary Information, Section 4.

The coordinated operation model uses detailed All-Island generation, demand and network data. Generator capacities, piecewise heat-rate curves and operating parameters are obtained from the Single Electricity Market model used in EirGrid publications [48]. Commodity costs and heat values are represented for non-gas fossil-fuel units. For gas-fired units, fuel use is linked to the gas-system module so that heat value

changes with hydrogen fraction. Onshore wind and solar generation are calculated from ERA5 wind speed and solar-radiation data; photovoltaic output is calculated using the Beckman model. Hydropower generation is taken from historical operation data [49].

Future gas demand is based on Gas Networks Ireland’s Gas Forecast Statement and reflects coal-plant retirements, the 80% renewable-generation ambition in Ireland’s Climate Action Plan, restrictions on new gas boiler connections, GDP-linked residential, industrial and commercial demand changes, and compressed natural gas station development [50]. Daily electricity peak demands to 2032 are taken from EirGrid’s Ten-Year Generation Capacity Statement for winter, summer and spring-autumn cases [51]. Missing values between 2033 and 2050 are filled using an autoregressive integrated moving average model. Fifteen-minute electricity and gas demand profiles are taken from EirGrid’s smart-grid dashboard and the Gas Networks Ireland data dashboard [52, 53]. High-voltage power lines above 110 kV and high-pressure gas pipelines above 16 bar are represented using network topology, location and parameter data from EirGrid and Gas Networks Ireland [54, 55]. Gas-pipeline friction factors and diameters follow an Ireland case study [56]; power-line capacities are assumed to grow at the same annual rate as demand.

4.4 International green hydrogen and ammonia trading model

After domestic absorption is accounted for, surplus offshore hydrogen and ammonia can be transported by ship to demand centres elsewhere in Europe. The international trade model uses the country-level cost-supply curves and available export potentials derived above to optimise hydrogen and ammonia production, shipping, domestic self-supply and external imports in 2030, 2040 and 2050. Countries with published offshore wind targets, including Ireland [57] and the UK [58], use those targets as maximum offshore capacities. For countries without complete year-specific targets, we use tentative goals from the European Union’s non-binding agreement on offshore wind deployment [59]. Hydrogen and ammonia demands are taken from the European Hydrogen Backbone demand projection [60].

The integrated model minimises total production, shipping and external-import costs subject to national offshore supply capacities, domestic export limits, country-level hydrogen and ammonia demand, shipping-route capacities and material balances. Decision variables include hydrogen and ammonia production by country, bilateral hydrogen and ammonia route flows, the number of hydrogen and ammonia vessels assigned to each route, and hydrogen and ammonia imports from the external import node. Hydrogen and ammonia production costs are represented through piecewise-linear interpolations of the cost-supply curves. Shipping costs include annualised vessel investment costs and operating fuel costs, with route distances derived from port-to-port shipping paths [19]. Because the supply curves and vessel-count decisions introduce discrete choices, the model is formulated as a mixed-integer linear programming problem and solved in MATLAB using Gurobi. The complete formulation is given in Supplementary Information, Section 5.

External imports are represented by an international import node. In the base integrated model, this node is retained so that demand can still be met when European offshore supply is insufficient. In the outside-option sensitivity, the same node is assigned a delivered hydrogen-equivalent price and the model is solved repeatedly over 2–6 EUR kg⁻¹ H₂-equivalent, corresponding to 60–180 EUR MWh⁻¹. Hydrogen and ammonia imports are assigned the same hydrogen-equivalent delivered price so that the scan isolates the threshold at which European offshore supply becomes competitive with non-European low-carbon imports.

Carbon mitigation is reported as a direct displacement-accounting metric rather than a full lifecycle assessment. Optimised hydrogen and ammonia deliveries are converted to avoided CO₂ using the carrier-specific conversion factors defined in the Supplementary Information and assigned to a supplier-destination contribution matrix. Positive bilateral route variables assign the avoided CO₂ to the exporting country and importing country; negative route variables reverse the direction before assignment. Domestic self-supply is assigned to the matrix diagonal, and the external import node is reported separately from the 11 European offshore supplier countries. The total carbon mitigation reported in the main text is the sum of the full matrix, while the European offshore supplier contribution excludes the external import row.

For the lowest-LCOH-only counterfactual, total European hydrogen and ammonia demand is allocated to offshore supply segments in ascending LCOH order. This baseline uses planned offshore capacity and capacity factors to limit country supply, but deliberately excludes domestic absorption limits, shipping costs, demand geography and the competitive external-import price. It is therefore used only as a diagnostic baseline for testing whether exporter rankings can be inferred from production cost alone.

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Author contributions

S.W. conceived the study, developed the modelling framework and MATLAB code, performed the analysis, prepared the figures and wrote the original manuscript. M.M.B. contributed to the energy-system interpretation, policy context and manuscript review and editing. Both authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Data availability

The public data sources used in this study are cited in the References and described in the Methods and Supplementary Information. Derived source-data tables generated by the analysis, including the headline-number audit table used to trace manuscript values and a source-data coverage inventory, are available in the project repository under `results/tables/`. Final figure files are provided under `manuscript/figs/`. Large public raw climate, bathymetry and geospatial inputs and large intermediate MATLAB checkpoint files are not redistributed through the GitHub repository because of file-size constraints; the raw inputs are available from the cited public data providers and the checkpoints can be regenerated from the provided code.

Code availability

MATLAB code used in this study is available on GitHub at <https://github.com/ShengWang-EE/HyExport>.

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